

are reaching out and holding an object.

There are several types of holographic displays. Of these, augmented holographic displays, which embed 3D objects into your surroundings, as if these were real, are already here. Products like Microsoft HoloLens and MagicLeap are examples of augmented holographic displays that provide an interactive, customisable experience.

In another class of holographic displays, nanophotonic devices manipulate light to achieve moving, full-colour holograms. These devices manipulate light at the source rather than trick the eye using middleware like glasses.

Several teams across the world, including one at Griffith University, Australia, are trying to develop such systems using graphene oxides and laser pulses to control the refractive index and wavelength of light. A nano-photonic holographic display does not require glasses, which means several people can view it. It is also potentially scalable to any size. However, it needs a wall to work and cannot project onto thin air.

If projecting onto thin air is what you really want, then you will get that, too. A team of researchers from Aerial Burton and multiple Japanese universities recently demonstrated a system that projects pixie-shape holograms called Fairy Lights into thin air. What makes these holograms all the more special is that these respond to human touch. The system uses femtosecond lasers, which can stimulate physical matter to emit light in 3D form.

Micro-LEDs to give OLEDs a good chase

Display experts consider micro-LEDs a technology worth watching. Micro-LED is an emerging display technology that uses arrays of microscopic LEDs as individual pixel elements. The tech offers benefits like greater contrast, faster response times and low energy consumption. While larger micro-LED displays are still too

difficult to manufacture, experts feel that these displays make sense for smaller devices like smartwatches. That explains why Apple acquired LuxVue, a company working on micro-LED displays.

That said, perhaps micro-LED might emerge in larger formats. When John Rogers and Chris Bower of X-Celeprint proposed the use of micro-LEDs for multifunctional display systems at a conference last year, they first described a process for making multiple layers of LEDs with sacrificial layers in between that allow the layers to be lifted off. The company has developed a tech-

If all this talk about holograms has made you thirsty for experimentation, try this not-so-little trick that turns your smartphone into a hologram projector: www.youtube.com/watch?v=7YWTtCsVgvg&feature=youtu.be

nology for transfer-printing of chips using elastomeric stamps utilising peel-rate-dependent adhesion.

If that does not make sense, let us turn to industry expert Kim Werner's simplified explanation on *Display Daily*: "To oversimplify shamelessly, if you place the stamp on the layer of chips and peel it off quickly, the chips adhere to the stamp. Impress the stamp on the target substrate and peel it off slowly, the chips adhere to the target. This is also impressive, but it still does not create LED arrays any larger than the original lattice-matched arrays.

"As it turns out, it is relatively simple to impose patterns on stamps that result in picking up every tenth, twentieth or nth LED before depositing these on the substrate. In this way, you can go from the dense array of the original wafer to a sparse array on the target substrate. In principal, this allows you to make μ -ILED displays of virtually any diagonal."

Using this technology, X-Celeprint has made 150mm stamps. According to Bower, making larger ones is just a matter of engineering, not science.

Nature-inspired display tech

The very concept of display is in many ways inspired by nature. No wonder scientists continue to observe nature—from the butterfly's wings to the woodcock's 360-degree vision—to develop better display systems.

In one recent venture, scientists from Harvard University and MIT learnt a few lessons from a type of mollusc called the blue-rayed limpet. Limpets have two optical structures within their shell. These are configured to reflect blue light while absorbing all other wavelengths of incoming light, resulting in a blue-striped appearance.

In an attempt to understand where the stripes came from, researchers decided to take a close look at the translucent shells using 2D and 3D structural analysis to reveal the 3D nanoarchitecture of the photonic structures embedded in the shells. The shells' top and bottom layers were relatively uniform, with dense stacks of calcium-carbonate platelets and thin organic structures.

As they got about 30 microns under the shell surface, they found regular plates of calcium-carbonate morphed into a multi-layered structure with regular spacing between layers, like a zigzag pattern.

This could potentially serve as a design guide for engineering colour-selective, controllable, transparent displays that require no internal light source. This technology can be used in windows and glasses, which need to be transparent and at the same time overlay AR content, like say, a map. In a news release, one of the researchers said, "We believe that the limpet's approach to displaying colour patterns in a translucent shell could serve as a starting point for developing such displays."

Everywhere we look, there is another creature that inspires a better display, another device that needs a better display and another team ready to work on it. Indeed, there is a lot to look forward to in the coming years. **EFY**